Teaching Statement

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When I started college, I had very little knowledge about diverse areas of Computer Science. At college, I took courses under some teachers who had extraordinary teaching capabilities, making a topic simple to the student and growing love for the subject among the students. The study of my undergraduate life prepared me for my next challenge in graduate life. After starting my Ph.D. study, I not only studied under great professors but also had the opportunity to teach undergraduate students. The opportunity to gain new knowledge and distribute it among the new generation of students is my primary motivation for joining the faculty position. If my teaching style makes an impact on my students' lives, I consider that as one of my greatest achievements.

Teaching Experience

During my Ph.D. life, I worked both as an Instructor and as a Graduate Teaching Assistant (GTA). As an Instructor, I taught the Computer Science I (Object-Oriented Programming) course for three semesters. I was solely responsible for conducting the full course, which included preparing course slides, delivering lectures, evaluating exam papers and assignments, and preparing exam questions. I always try to make the topic easier to understand by telling a story and by providing real-world examples. I have also worked as a GTA for a number of courses. Among them, the Senior Capstone Project was particularly meaningful because it gave me the opportunity to mentor students as they applied their technical knowledge to real-world projects. In this course, final-year undergraduate students are assigned to group projects for developing professional software. Often, tech companies (such as General Motors) propose project ideas, and students work on completing the project. I have worked closely with the course instructor to design the course materials and course structure. Students are then mentored by GTAs to ensure the successful completion of the project. As those projects were proposed by software companies, I mentored the students in a way so that they could deliver a professional software product. The students were responsible for preparing documentation of the software, following standard software development practices, providing updates to the clients in each sprint, and delivering a professional software product to the client.

Because of working as an Instructor and GTA, I have interacted with many students and software industry professionals. My teaching experience has also helped me learn about course material design, delivering lectures, assignment design, and grading policies. Students feel comfortable interacting with me and asking me questions related to the course topics. I also enjoy interacting with students and teaching them new topics.

Teaching Philosophy

I have seen throughout my teaching career that students learn better when they are taught by practical examples. Even when I explain a theoretical concept, I first explain the implication of that theory in the real-world scenario, then describe the theory, and finally provide real examples that can help to understand the theory well. I view the role of an educator as a mentor who guides students on a journey of self-discovery and personal growth. Each student has a unique learning style, and an educator needs to recognize the style and guide the students accordingly. I always believe in creating a supportive environment in the classroom that would propel the students to navigate the world's complexities. By enforcing a sense of self-confidence, my goal is to help the students become independent thinkers. I prefer to incorporate technologies to

enhance the learning experience of the students. I not only prioritize academic learning but also emphasize developing personal skills such as collaboration, problem-solving, and communication. Education is not just acquiring knowledge but understanding the strengths of a person so that he/she can utilize it for the betterment of society.

I try to introduce the students to cutting-edge research techniques to stimulate their critical thinking. Because of the evolving nature of the software engineering field, I am committed to ensuring a dynamic learning environment. As different students have different learning capabilities, I try to address the needs of individual students according to their ability to learn. To make the class more interactive, I asked the students questions after finishing each topic. Those questions often lead to discussion, which helps the students clearly understand that topic. At Wayne State University, I have taught and mentored students from ethnically and socio-economically diverse backgrounds. As I embark on my future in teaching, my dedication extends to collaborating with students of varied backgrounds.

My Future Courses

Considering my experience as an instructor in academia and my research experience in Bioinformatics and Deep learning domain, I am interested in designing and conducting a variety of courses at the undergraduate and graduate levels.

- **Machine learning:** Machine learning is a fundamental area of computer science that enables computers to identify patterns in data and make predictions or decisions without being explicitly programmed for every task. I intend to design the course in a way so that students develop both a strong understanding of the underlying theory and practical experience applying machine learning methods to real-world datasets. The course will cover essential topics such as data preprocessing, feature selection, supervised and unsupervised learning, classification, regression, clustering, ensemble methods, model evaluation, overfitting, and regularization. Students will also learn how to select appropriate algorithms, evaluate model performance, and interpret the results critically. I plan to include programming assignments and a semester project in which students will build, test, and compare machine learning models using real-world data. Through these activities, students will strengthen their analytical, algorithmic, and implementation skills of machine learning models.
- **Deep learning:** Learning traditional machine learning methods alone is not enough; students also need to understand how modern neural networks learn complex patterns from large and high-dimensional datasets. In this course, I will teach students how to design, train, evaluate, and interpret models such as feedforward neural networks, convolutional neural networks, recurrent neural networks, autoencoders, graph neural networks, and generative adversarial networks. As I have experience conducting research using deep learning for biomedical and genomic applications, I would show students how these methods can be applied to real-world scientific problems. As this field continues to evolve rapidly, I will provide students with a selection of both foundational and recently published research papers. Studying foundational papers will help them understand the core ideas that have shaped modern deep learning, while recent papers will keep them informed about the latest advances in deep learning and large language models (LLMs). I believe that developing the ability to read and critically analyze research papers is an essential skill for students who plan to pursue graduate studies or a Ph.D.
- **Bioinformatics:** Modern biology has increasingly become a data-driven field, making the ability to retrieve, analyze, and interpret high-dimensional biological data valuable for students interested in biomedical research. In this course, I will introduce students to R-based bioinformatics, reproducible data analysis, visualization, statistical reasoning, transcriptomic analysis, genome browsing, and multi-omics integration. I will guide students gradually from fundamental concepts to practical analysis. Drawing on my Ph.D. research in cancer genomics and computational biology, I will use real biomedical examples

and an independent project to help students develop skills relevant to graduate study, biotechnology, and data-intensive biological research.

- **Object-oriented programming:** I have experience conducting this course at Wayne State University and I am very much interested in teaching this course in my future teaching career. Object-oriented programming is the foundation of modern software development. My experience teaching this course for three semesters has made me comfortable not only explaining object-oriented concepts but also demonstrating them through practical programming examples and real-world applications. I plan to discuss object-oriented design principles, classes and objects, encapsulation, inheritance, polymorphism, abstraction, exception handling, file processing, and fundamental data structures, which I believe will help students develop the programming and problem-solving skills necessary for designing robust, maintainable, and scalable software systems.
- **Other Courses:** I am also interested in teaching a variety of courses, including software engineering, programming languages, algorithm design, operating system, theory of computation, and senior capstone project. As the computer science field is evolving quickly, I am prepared to design and teach new courses at the undergraduate and graduate levels. Considering my experience as an educator, and as a researcher, I am confident that I will be able to teach most courses in the computer science field.

Experience of Mentoring Students

Throughout my Ph.D. career, I have mentored a number of students in research and academic courses. As part of the Senior Capstone Project course, I have mentored 48 final-year undergraduate students, where students developed professional applications for software companies. I have worked with them closely and mentored them to follow standard practices of the software industry. I have also worked with three undergraduate and master's students on different research projects. I prefer to guide them in a way that they can discover their own strength rather than imposing something on my mentees.

I agree with the philosophy of Horace Mann, who said, "A teacher who is attempting to teach without inspiring the pupil with a desire to learn is hammering on cold iron." Mentoring a student means finding that student's unique strengths and nurturing those strengths, which would empower that person in his/her future journey. I make the students aware of their goals and help them so that they can find their own way of achieving that goal. By mentoring my students, I often learn new ideas from them. I believe a friendly relationship between the mentor and the mentee is a catalyst for achieving common goals. I always remind my mentees about the importance of time management and setting smaller goals, which in turn results in the successful completion of the project.